PADHAKU
Science · Class 10 Science



CHAPTER 4

Carbon and its Compounds

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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Activity 4.1

1 **Make a list of ten things you used since morning and sort them into a table:**

Made of metal	Made of glass/clay	Others

💡 **SAMPLE ANSWER**

Sample table:

Made of metal	Made of glass/clay	Others
spoon	cup	toothbrush (plastic)
bucket	plate	soap
pen nib	tumbler	clothes, paper, food, wooden chair

Most items in the **'Others'** column (food, clothes, paper, plastic, wood, soap, medicines) are made of **compounds of carbon**.

? **Can you think of a method to test that these contain carbon? What is the product when a carbon compound is burnt?**

✔ **ANSWER**

Burn the substance. A carbon compound gives **carbon dioxide (CO_2)** on burning.

Test: pass the gas through lime water – it turns **milky**, which confirms CO_2, so carbon was present.

🤔 Think · 4.1 Bonding in Carbon – The Covalent Bond

? **Atomic number of carbon is 6. Distribution of electrons? How many valence electrons?**

✔ ANSWER

K = 2, L = 4, i.e. **2, 4**. Carbon has **4 valence electrons**.

? **Chlorine (atomic number 17): electronic configuration, valency and electron dot structure of Cl_2.**

✔ ANSWER

Configuration **2, 8, 7**; valency **1** (needs 1 electron).

Cl_2: each Cl shares one electron → **one shared pair = single bond**, Cl–Cl.

Each Cl also keeps **3 lone pairs** (6 dots), so each gets an octet.

? **Depict a molecule of water. Does it have single or double bonds?**

✔ ANSWER

O (2, 6) shares one electron pair with each of the two H atoms: **H–O–H**.

O also has **2 lone pairs**; O gets an octet, each H gets 2 electrons.

It has **two single bonds**.

? **Nitrogen (atomic number 7): electronic configuration and combining capacity?**

✔ ANSWER

Configuration **2, 5**; combining capacity (valency) **3**.

In N_2 each N shares 3 electrons → **3 shared pairs = triple bond**, N≡N; each N has 1 lone pair.

? **Draw the electron dot structure of ammonia, NH_3. Single, double or triple bonds?**

✔ ANSWER

N shares one electron pair with each of the 3 H atoms; N keeps **1 lone pair**.

N gets an octet, each H gets 2 electrons (like helium).

All are **single bonds** (3 N–H bonds).

? Questions · 4.1 Bonding in Carbon – The Covalent Bond

Q 1 **What would be the electron dot structure of carbon dioxide which has the formula CO_2?**

✔ ANSWER

C has 4 valence electrons, each O has 6.

C shares **two electron pairs with each O** → two double bonds.

Dot structure: **O::C::O**, each O also has **2 lone pairs**.

Structural formula: **O=C=O**. Now C and both O atoms have an octet.

Q 2 **What would be the electron dot structure of a molecule of sulphur made up of eight atoms of sulphur (S_8)? (The atoms form a ring.)**

✔ ANSWER

S (2, 8, 6) has 6 valence electrons and needs 2 more.

The 8 S atoms form a **crown-shaped ring**; each S shares **one electron pair with each of its two neighbours** (two single bonds).

Each S also keeps **2 lone pairs**, so every S has an octet.

Total: **8 S–S single bonds** in the ring.

🤔 Think · 4.2 Versatile Nature of Carbon

? **Draw the structure of propane, C_3H_8.**

✔ ANSWER

Step 1: C–C–C. Step 2: fill valencies with H:

CH_3–CH_2–CH_3 (end carbons 3 H each, middle carbon 2 H).

All bonds are single (2 C–C + 8 C–H = 10 bonds) → saturated.

? **Draw the electron dot structure of ethyne, C_2H_2. How many bonds are needed between the two carbon atoms?**

✔ ANSWER

Each C shares one pair with an H and **three pairs with the other C**.

H–C≡C–H: a **triple bond** (3 shared pairs) is needed between the carbons.

? **Can you draw the electron dot structure for cyclohexane, C_6H_{12}?**

✔ ANSWER

6 C atoms form a ring; each C shares one pair with each neighbouring C and one pair with each of its **2 H** atoms.

All single bonds: **6 C–C + 12 C–H = 18 shared pairs**.

? **Difference between propane and butane (C_4H_{10})? Difference in molecular masses between these pairs? (C = 12 u, H = 1 u)**

✔ ANSWER

C_3H_8 and C_4H_{10} also differ by a **$-CH_2-$ unit**.

Mass of CH_2 = 12 + 2 = **14 u**: CH_4 16 u, C_2H_6 30 u, C_3H_8 44 u, C_4H_{10} 58 u – each pair differs by 14 u.

? Formula of ethene? Do C_2H_4, C_3H_6, C_4H_8, C_5H_{10} differ by $-CH_2-$? Relation between C and H? General formula for alkanes and alkynes?

✔ ANSWER

Ethene: **C_2H_4**. Yes, successive alkenes differ by $-CH_2-$.

Number of H = 2 × number of C → C_nH_{2n}.

Alkanes: **C_nH_{2n+2}** (n = 1, 2, 3...).

Alkynes: **C_nH_{2n-2}** (n = 2, 3, 4...).

Activity 4.2

1 Calculate the difference in formulae and molecular masses for (a) CH_3OH and C_2H_5OH (b) C_2H_5OH and C_3H_7OH (c) C_3H_7OH and C_4H_9OH.

✔ ANSWER

Masses: CH_3OH = 32 u, C_2H_5OH = 46 u, C_3H_7OH = 60 u, C_4H_9OH = 74 u.

(a) difference **$-CH_2-$**, 46 − 32 = **14 u**

(b) difference **$-CH_2-$**, 60 − 46 = **14 u**

(c) difference **$-CH_2-$**, 74 − 60 = **14 u**

2 **Is there any similarity in these three?**

✔ ANSWER

Yes. Each pair differs by one **$-CH_2-$ unit** and by **14 u** in mass.

3 **Arrange these alcohols in increasing carbon atoms. Can we call this family a homologous series?**

✔ ANSWER

CH_3OH (methanol) → C_2H_5OH (ethanol) → C_3H_7OH (propanol) → C_4H_9OH (butanol).

Yes, it is a homologous series: same functional group (–OH), successive members differ by $-CH_2-$, similar chemical properties.

4 **Generate the homologous series up to four carbons for the other functional groups in Table 4.3.**

✔ ANSWER

- Haloalkanes: CH_3Cl, C_2H_5Cl, C_3H_7Cl, C_4H_9Cl
- Aldehydes: HCHO (methanal), CH_3CHO (ethanal), CH_3CH_2CHO (propanal), $CH_3CH_2CH_2CHO$ (butanal)
- Ketones (start with 3 C): CH_3COCH_3 (propanone), $CH_3COCH_2CH_3$ (butanone)

- Carboxylic acids: HCOOH (methanoic), CH_3COOH (ethanoic), CH_3CH_2COOH (propanoic), $CH_3CH_2CH_2COOH$ (butanoic acid)

❓ Questions · 4.2 Versatile Nature of Carbon

Q 1 **How many structural isomers can you draw for pentane?**

✔ ANSWER

Three isomers of C_5H_{12}:

- n-pentane: $CH_3–CH_2–CH_2–CH_2–CH_3$ (straight chain)
- isopentane (2-methylbutane): $CH_3–CH(CH_3)–CH_2–CH_3$
- neopentane (2,2-dimethylpropane): $C(CH_3)_4$

Q 2 **What are the two properties of carbon which lead to the huge number of carbon compounds we see around us?**

✔ ANSWER

- **Catenation**: carbon can bond with other carbon atoms to form long chains, branched chains and rings.
- **Tetravalency**: carbon has valency 4, so it can bond with four other atoms (C, H, O, N, S, Cl...).

Q 3 **What will be the formula and electron dot structure of cyclopentane?**

✔ ANSWER

Formula: **C_5H_{10}**.

5 C atoms form a ring; each C shares one pair with each of its 2 neighbouring C atoms and one pair with each of its **2 H** atoms.

All single bonds: **5 C–C + 10 C–H = 15 shared pairs**; each C has an octet.

Q 4 **Draw the structures for: (i) Ethanoic acid (ii) Bromopentane (iii) Butanone (iv) Hexanal. Are structural isomers possible for bromopentane?**

✔ ANSWER

(i) Ethanoic acid: **$CH_3–COOH$**, i.e. $H_3C–C(=O)–O–H$

(ii) Bromopentane: **$CH_3–CH_2–CH_2–CH_2–CH_2–Br$**

(iii) Butanone: **$CH_3–CO–CH_2–CH_3$**, i.e. $H_3C–C(=O)–CH_2–CH_3$

(iv) Hexanal: **$CH_3–CH_2–CH_2–CH_2–CH_2–CHO$**

Yes, bromopentane has structural isomers, e.g. 1-bromopentane (Br on end carbon), 2-bromopentane $CH_3–CHBr–CH_2–CH_2–CH_3$, 3-bromopentane $CH_3–CH_2–CHBr–CH_2–CH_3$, and others with branched chains.

Q 5 How would you name the following compounds?

(i) $CH_3–CH_2–Br$

(ii) $H–C(H)=O$ (HCHO)

(iii) $H–C(H_2)–C(H_2)–C(H_2)–C(H_2)–C\equiv C–H$ ($CH_3CH_2CH_2CH_2C\equiv CH$)

✔ ANSWER

(i) **Bromoethane**

(ii) **Methanal** (formaldehyde)

(iii) **Hexyne** (1-hexyne): 6 carbons with a triple bond

🤔 Think · 4.3.1 Combustion

? Balance: $CH_4 + O_2 \rightarrow CO_2 + H_2O$ and $CH_3CH_2OH + O_2 \rightarrow CO_2 + H_2O$.

✔ ANSWER

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat and light

$CH_3CH_2OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$ + heat and light

🧪 Activity 4.3

1 Burn naphthalene, camphor and alcohol one by one. Note the flame and smoke. Is there a deposit on a metal plate held above?

✔ ANSWER

Naphthalene and camphor burn with a **yellow, sooty flame** and give black smoke; a **black soot deposit** forms on the plate.

Alcohol burns with a **clean blue flame**, no smoke and no deposit.

Conclusion: unsaturated compounds (more carbon) burn incompletely with a sooty flame; saturated ones burn with a clean flame.

Activity 4.4

1 **Adjust the air hole of a bunsen burner. When do you get a yellow, sooty flame? When do you get a blue flame?**

✔ ANSWER

Air hole closed (less air) → incomplete combustion → **yellow, sooty flame**.

Air hole open (enough air) → complete combustion → **blue flame**.

Do You Know? / More to Know!

? **Incomplete combustion gives soot (carbon). What will you attribute the yellow colour of a candle flame to?**

✔ ANSWER

To **unburnt carbon (soot) particles** that get heated and glow, giving the yellow colour.

? **Can you guess why coal and petroleum are called fossil fuels?**

✔ ANSWER

They were formed over millions of years from the **remains (fossils) of plants and animals** buried under the earth, so they are called fossil fuels.

Activity 4.5

1 **Warm 3 mL ethanol and add 5% alkaline potassium permanganate drop by drop. Does the colour persist at first? Why does it not disappear when excess is added?**

✔ ANSWER

At first the **purple colour disappears**: $KMnO_4$ is used up in oxidising ethanol to ethanoic acid.

CH_3CH_2OH —(alkaline $KMnO_4$ + heat)→ CH_3COOH

When **excess** is added, all ethanol is already oxidised, so no more $KMnO_4$ is used and the **colour stays**.

Questions · 4.3 Chemical Properties of Carbon Compounds

Q 1 **Why is the conversion of ethanol to ethanoic acid an oxidation reaction?**

✔ ANSWER

Oxygen is **added** (and hydrogen removed) when ethanol changes to ethanoic acid:

CH_3CH_2OH —(alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$, heat)→ CH_3COOH

Addition of oxygen is oxidation, so it is an **oxidation reaction**.

Q 2 **A mixture of oxygen and ethyne is burnt for welding. Why is a mixture of ethyne and air not used?**

✔ ANSWER

Air has only about 21% oxygen. Ethyne burns **incompletely** in air, giving a sooty flame and **less heat**.

With pure oxygen, ethyne burns **completely** and gives a very high temperature (about 3000 °C), which is needed to **melt metals** for welding.

Activity 4.6

1 **Drop a small piece of sodium into ethanol (absolute alcohol). What do you observe?**

How will you test the gas evolved?

✔ ANSWER

Bubbles of gas come out (effervescence) and sodium slowly disappears.

$2Na + 2CH_3CH_2OH \rightarrow 2CH_3CH_2O^-Na^+$ (sodium ethoxide) + H_2

Test: bring a **burning splinter** near the gas – it burns with a **'pop' sound** → hydrogen.

? **Can you recall which other substances produce hydrogen on reacting with metals?**

✔ ANSWER

Acids (e.g. dilute HCl, dilute H_2SO_4) with active metals like zinc, and **water** with very reactive metals like sodium.

Activity 4.7

1 **Compare dilute acetic acid and dilute hydrochloric acid using litmus and universal indicator. Are both acids shown by litmus? Does the universal indicator show them equally strong?**

✔ ANSWER

Yes, both turn **blue litmus red**.

No. Universal indicator shows a **lower pH for HCl** (about 1–2, red) and a **higher pH for acetic acid** (about 3–4, orange).

So acetic acid is a **weak acid** (only partly ionised) while HCl is a strong acid.

Activity 4.8

1 **Warm 1 mL ethanol + 1 mL glacial acetic acid + a few drops of conc. sulphuric acid; pour into water and smell.**

✔ ANSWER

A **sweet, fruity smell** is noticed – an **ester** (ethyl ethanoate) is formed.

$CH_3COOH + CH_3CH_2OH \xrightarrow{(acid)} CH_3COOCH_2CH_3 + H_2O$

This is called **esterification**.

Activity 4.9

1 **Add dilute ethanoic acid to sodium carbonate; pass the gas through lime water. What do you observe? Can the gas be identified? Repeat with sodium hydrogencarbonate.**

✔ ANSWER

Brisk effervescence (bubbles) is seen.

The gas turns **lime water milky** → it is **carbon dioxide**. Yes, it is identified by this test.

$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$

With sodium hydrogencarbonate the **same result**: $CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2$

❓ Questions · 4.4 Ethanol and Ethanoic Acid

Q 1 **How would you distinguish experimentally between an alcohol and a carboxylic acid?**

✔ ANSWER

Add a pinch of **sodium hydrogencarbonate** (or sodium carbonate) to each:

- **Carboxylic acid**: brisk effervescence of **CO_2**, which turns lime water milky.
- **Alcohol**: **no reaction**.

(Also: a carboxylic acid turns blue litmus red; an alcohol does not.)

Q 2 **What are oxidising agents?**

✔ ANSWER

Substances that **add oxygen** to other substances (or remove hydrogen) are oxidising agents.

Examples: **alkaline potassium permanganate** and **acidified potassium dichromate** oxidise ethanol to ethanoic acid.

Activity 4.10

1 **Shake oil + water in tube A, and oil + water + soap solution in tube B. Can you see separate layers just after shaking? Does the oil layer separate on standing? In which tube first?**

✔ ANSWER

Just after shaking: in **A** oil and water layers start separating quickly; in **B** the oil is spread as tiny droplets (an **emulsion**), so no separate layer is seen.

On standing, the oil layer separates **first in tube A** (no soap). In B it stays mixed for a long time.

Conclusion: soap forms **micelles** around oil and keeps it in water – this is how soap cleans.

? **Can you draw the structure of the micelle formed if you dissolve soap in a hydrocarbon?**

✔ ANSWER

It is an **inverted (reverse) micelle**: the **ionic (hydrophilic) heads** are inside, at the centre, and the **hydrocarbon tails** point outward into the hydrocarbon solvent.

Activity 4.11

1 **Add soap solution to distilled water and to hard water and shake. In which tube do you get more foam? In which do you see a white curdy precipitate?**

✔ ANSWER

More foam in distilled (or rain) water.

White curdy precipitate (scum) in hard water: Ca^{2+} and Mg^{2+} ions react with soap to form insoluble salts.

Activity 4.12

1 **Add soap solution to one tube of hard water and detergent solution to another. Same amount of foam? In which tube is a curdy solid formed?**

✔ ANSWER

No. The **detergent** tube gives **more foam**.

The curdy solid (scum) forms in the **soap** tube. Detergents do not form insoluble precipitates with Ca^{2+}/Mg^{2+}, so they work in hard water.

❓ Questions · 4.5 Soaps and Detergents

Q 1 **Would you be able to check if water is hard by using a detergent?**

✔ ANSWER

No. A detergent gives good foam in both hard and soft water and forms no scum, so it cannot show hardness.

Soap is used instead: hard water gives little foam and a curdy scum.

Q 2 **Why is agitation (beating, scrubbing, washing machine) necessary to get clean clothes?**

✔ ANSWER

Soap micelles trap oily dirt, but the dirt is stuck to the cloth fibres.

Beating, scrubbing or agitation **loosens the dirt** from the fibres and helps more micelles form around it, so the dirt goes into the water and is **rinsed away**.

📝 Exercises

1 **Ethane, with the molecular formula C_2H_6 has (a) 6 covalent bonds (b) 7 covalent bonds (c) 8 covalent bonds (d) 9 covalent bonds**

✔ ANSWER

(b) 7 covalent bonds – 1 C–C bond + 6 C–H bonds.

2 **Butanone is a four-carbon compound with the functional group**

(a) carboxylic acid (b) aldehyde (c) ketone (d) alcohol

✔ ANSWER

(c) ketone – the suffix '-one' shows a ketone group (>C=O).

3 **While cooking, if the bottom of the vessel is getting blackened on the outside, it means that**

(a) the food is not cooked completely (b) the fuel is not burning completely (c) the fuel is wet (d) the fuel is burning completely

✔ ANSWER

(b) the fuel is not burning completely – incomplete combustion leaves soot (carbon); the air holes are blocked.

4 **Explain the nature of the covalent bond using the bond formation in CH_3Cl.**

✔ ANSWER

C (2, 4) has 4 valence electrons, H has 1, Cl (2, 8, 7) has 7.

C shares **one electron pair with each of 3 H atoms** and **one pair with the Cl atom** → 4 single covalent bonds.

Now C and Cl have an octet and each H has 2 electrons (noble gas configuration).

A covalent bond is formed by **sharing an electron pair**; no ions form. Bonds inside the molecule are strong, but forces between molecules are weak, so such compounds have low melting/boiling points and do not conduct electricity.

5 **Draw the electron dot structures for (a) ethanoic acid (b) H_2S (c) propanone (d) F_2.**

✔ ANSWER

(a) **Ethanoic acid, CH_3COOH** ($H_3C–C(=O)–O–H$): the CH_3 carbon shares one pair with each of 3 H and one pair with the second C; the second C shares **two pairs with one O** (C=O) and one pair with the other O; that O shares one pair with H. Each O has **2 lone pairs**.

(b) **H_2S** (H–S–H): S (2, 8, 6) shares one pair with each H; S keeps **2 lone pairs**.

(c) **Propanone, CH_3COCH_3** ($H_3C–C(=O)–CH_3$): middle C shares one pair with each CH_3 carbon and **two pairs with O** (C=O); each end C shares one pair with each of its 3 H; O has **2 lone pairs**.

(d) **F_2** (F–F): each F (2, 7) shares one electron → **one shared pair**; each F keeps **3 lone pairs**.

6 **What is a homologous series? Explain with an example.**

✔ ANSWER

A **homologous series** is a series of carbon compounds with the **same functional group** (or same general formula), in which successive members differ by a **$-CH_2-$ unit** (14 u).

Members have **similar chemical properties**, while physical properties (melting point, boiling point, solubility) change gradually with molecular mass.

Example – alkanes (C_nH_{2n+2}): CH_4 (methane), C_2H_6 (ethane), C_3H_8 (propane), C_4H_{10} (butane).

Another: alcohols CH_3OH, C_2H_5OH, C_3H_7OH.

7 **How can ethanol and ethanoic acid be differentiated on the basis of their physical and chemical properties?**

✔ ANSWER

Property	Ethanol	Ethanoic acid
Melting point	156 K	290 K (freezes in winter)
Boiling point	351 K	391 K
Smell	pleasant (spirit-like)	vinegar-like

Litmus	no change	blue litmus turns red
With $NaHCO_3$ / Na_2CO_3	no reaction	brisk effervescence of CO_2
With NaOH	no reaction	forms salt + water ($CH_3COONa + H_2O$)

8 **Why does micelle formation take place when soap is added to water? Will a micelle be formed in other solvents such as ethanol also?**

✔ ANSWER

A soap molecule has an **ionic (hydrophilic) end** that dissolves in water and a long **hydrocarbon (hydrophobic) tail** that does not.

In water the tails keep away from water and **cluster together inside**, while the ionic ends face the water → a **micelle**.

No, micelles do not form in ethanol: the hydrocarbon tail is soluble in ethanol, so the tails need not cluster together.

9 **Why are carbon and its compounds used as fuels for most applications?**

✔ ANSWER

- They burn in air (oxygen) giving out a **large amount of heat and light**.
- They have a **high calorific value** and burn easily at a controlled rate.

- Most are **easily available**, cheap and easy to store and carry.
- Saturated hydrocarbons burn with a clean flame, giving mainly CO_2 and water.

10 **Explain the formation of scum when hard water is treated with soap.**

✔ ANSWER

Hard water contains **calcium and magnesium salts** (Ca^{2+}, Mg^{2+} ions).

Soap is a sodium/potassium salt of a long-chain carboxylic acid. It reacts with these ions to form **insoluble calcium and magnesium salts**, which separate as a white curdy solid – **scum**.

e.g. $2C_{17}H_{35}COONa + Ca^{2+} \rightarrow (C_{17}H_{35}COO)_2Ca\downarrow + 2Na^+$

So more soap is wasted and less foam forms.

11 **What change will you observe if you test soap with litmus paper (red and blue)?**

✔ ANSWER

Soap solution is **basic (alkaline)**.

Red litmus turns blue; blue litmus shows **no change**.

12 **What is hydrogenation? What is its industrial application?**

✔ ANSWER

Hydrogenation is the addition of hydrogen to an unsaturated hydrocarbon in the presence of a catalyst like **nickel or palladium**, giving a saturated compound.

e.g. $CH_2=CH_2 + H_2 \xrightarrow{(Ni)} CH_3–CH_3$

Industrial use: **hydrogenation of vegetable oils** (with nickel catalyst) to make **vanaspati ghee** (solid fat).

13 **Which of the following hydrocarbons undergo addition reactions: C_2H_6, C_3H_8, C_3H_6, C_2H_2 and CH_4?**

✔ ANSWER

C_3H_6 (propene) and **C_2H_2 (ethyne)** – they are unsaturated (double/triple bond).

C_2H_6, C_3H_8 and CH_4 are saturated, so they do not undergo addition.

14 **Give a test that can be used to differentiate between saturated and unsaturated hydrocarbons.**

✔ ANSWER

Burning test: burn each in air.

- Saturated hydrocarbon → **clean blue flame**, little or no smoke.

- Unsaturated hydrocarbon → **yellow, sooty flame** with black smoke.

(Another test: unsaturated hydrocarbons **decolourise bromine water**; saturated ones do not.)

15 **Explain the mechanism of the cleaning action of soaps.**

✔ ANSWER

- A soap molecule has a **hydrophilic ionic head** ($-COO^-Na^+$) and a long **hydrophobic hydrocarbon tail**.
- Most dirt is oily and does not dissolve in water.
- In soapy water, the tails attach to the oily dirt and the ionic heads face the water, forming **micelles** with the dirt trapped at the centre.
- This forms an **emulsion**; micelles do not come together because the ionic surfaces repel each other.
- Rubbing/agitation loosens the dirt, and the micelles carrying dirt are **washed away** with water, leaving the cloth clean.

🎉 **Chapter 4 done! Revise the answers once more tomorrow.**